

Supplementary file

This document presents some additional concerns surrounding the a priori power analysis of sports and exercise science researchers.

1. Example of extremely large effect size

Figure 1 illustrates that some a priori power analyses report extremely large effect sizes. However, this does not necessarily imply such large effect sizes are implausible. Certain manipulations can yield very large effect sizes. For example, consider the Cohen's d of 3.18 reported in the study of Takei et al. (2021), which was obtained from a within-subject study examining the effect of three hypoxic conditions on power output during 4 x 30-second all-out efforts (Kon et al., 2015; Figure S1). This large effect size is plausible as a large decrease in power output over consecutive all-out efforts should be expected. However, whether this effect size was a reasonable justification of the effect size of interest is questionable. The authors aimed to determine whether acute moderate hypoxia, compared to normoxia, had no detrimental effect on power output during 3 x 30-s all-out efforts. To estimate the effect of hypoxia, a direct comparison between the hypoxic condition(s) and the normoxic condition would be necessary. Yet, the calculated Cohen's d of 3.18 corresponds to the effect of all-out efforts¹, not to the effect of hypoxia. The effect of hypoxia would be much smaller than the effect of the all-out efforts since it would correspond to the change in power output from bout 1 to the subsequent bouts between the the hypoxic and normoxic conditions.

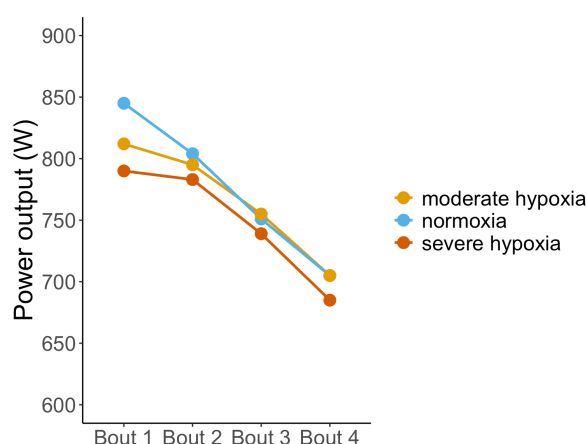


Figure S1. Pattern of means reported across 4 bouts and experimental condition. Figure S1 has been created using the means reported in Table 1 of the original study (Kon et al., 2015), from which the effect size (3.18) was based on.

Another issue is the lack of adequate justification for the choices researchers make during a priori power analyses. For instance, one study used a Cohen's d_z of 0.94 for its a priori power analysis, justifying this choice based on a

¹ "The sample size was estimated using a power analysis software G*Power (version 3.1.9.2; Bonn University, Bonn, Germany) based on the mean effect ($d = 3.18$) of the within-condition decrement to peak and mean power output during repeated 30-s Wingate efforts.¹⁰ The power analysis resulted in a calculated total sample size of 4 participants."

previous study examining the effect of omitting breakfast on evening endurance performance despite complete dietary compensation at lunch. However, such a large effect size appears implausible given that the overall energy intake and macronutrient profile was identical between conditions. Furthermore, the study from which the effect size was calculated employed a within-subject design and reported that the time taken to complete the time trial in the breakfast trial (465.7 ± 43.3 s) was shorter compared with the breakfast omission condition (469.2 ± 43.4 s). The original study does not report t -statistics so it is likely that authors used equation (8) from (Lakens, 2013) which requires the correlation between measurements to compute Cohen's d_z . To obtain the reported Cohen's d_z of 0.94 using the reported means and SDs requires a correlation of 0.99632 between measurements. The correlation estimate heavily influences the estimated Cohen's d_z . For example, assuming a correlation of 0.8 and using the same means and SDs, equation (8) returns Cohen's d_z of 0.13. An a priori power analysis intended to detect an effect size of $d_z = 0.13$, with a desired statistical power of 80% and a type I error of $\alpha = 0.05$ requires a total sample size of 467 participants as opposed to the sample size of 11 reported in the study in question. The correlation used to determine a sample size of 11 seems highly unrealistic.

2. Using a Cohen's d from a previous study to conduct an a priori power analysis for an interaction effect

2.1. Two between-subject factorial design

Researchers often conduct studies to test whether an effect is moderated by a second factor that suppresses, exacerbates or reverses the effect caused by a first factor— what is known as an interaction effect—. In our experience, many researchers assume it is reasonable to select an effect size from a previous study that employed a simple design (i.e., one or two groups)— note this also applies to factorial design with one of the factors with two levels ($df_1 = 1$)— to conduct an a priori power analysis for an interaction effect. However, whether the selected effect size is a good estimate of the interaction effect depends on the type of interaction or pattern of means that researchers expect to observe. Broadly speaking, there are three types of interactions: ordinal, disordinal or attenuated interaction (Figure S1). Readers are referred to Sommet et al. (2023) for a discussion about how different types of interactions affect power. To understand how the patterns of means affect the interaction effect size, it helps to conceive interactions as a “difference of differences” (Langenberg et al., 2023; Sommet et al., 2023). Under this approach, the calculation of a two-way interaction corresponds to the difference between a) the difference between the two levels of a factor at one level of the second factor and b) the difference between the two levels of a factor at the second level of the second factor. The advantage of this approach is that researchers can conduct an a priori power analysis for an interaction effect using a t -test and solely requires the specification of an expected effect size in terms of Cohen's d . For a 2×2 between-subject design, the interaction effect can be calculated as follows (Sommet et al., 2023):

$$d_{int} = \frac{(E_1 - C_1)}{Pooled\ SD} - \frac{(E_2 - C_2)}{Pooled\ SD} = \frac{(E_1 - C_1) - (E_2 - C_2)}{2 \times Pooled\ SD} \quad (3)$$

Where E and C refer to the experimental and control group respectively, and the subscripted numbers refer to the levels of the second factor. Thus, d_{int} boils down to computing the difference between two standardised mean differences (d_s) from the two main effects for which an a priori power analysis can be conducted in the framework of an unpaired t -test. To see how the pattern of means affects the interaction effect size, imagine we wish to test

whether the effect of a supplement is moderated by the intake time (i.e., morning vs. afternoon) using a 2 x 2 between-subject design. Specifically, we expect that taking the supplement in the evening will knock out the effect of taking the supplement in the morning—what is known as an ordinal interaction (Figure S1d). To conduct the a priori power analysis for this interaction effect, we rely on a previous study based on a between-subject design with two groups that found that the supplement of interest improved cycling time to exhaustion in comparison to a placebo. Assuming an equal standard deviation of 2, the improvement in time to exhaustion would correspond to a Cohen's $d_s = 0.5$ ($(1 - 0) / 2 = 0.5$; Figure S2a). Now imagine that a researcher uses this Cohen's d_s as an estimate of the hypothesized ordinal interaction to conduct the a priori power analysis. Plugging an effect size of $d_s = 0.5$, a type I error of $\alpha = 0.05$, and a desired statistical power of 0.8 into G*Power yields a total sample size of 128 for a two-sided unpaired t -test, or 32 participants for each of the four groups in our 2 x 2 factorial design. However, this a priori power analysis would be invalid because the chosen effect size does not correspond to the hypothesized ordinal interaction. For the hypothesized ordinal interaction (Figure S1b), the pattern of means would correspond to a $d_{int} = 0.25$ ($((1 - 0) - (0 - 0)) / (2 \times 2) = 0.25$). Plugging an effect size of $d_{int} = 0.25$, a type I error of $\alpha = 0.05$, and a desired level of statistical power to 0.8 into G*Power yields a total sample size of 506 for a two-sided unpaired t -test, or about 127 participants per group. For a reverse interaction (Figure S2c), the pattern of means would correspond to a $d_{int} = 0.5$ ($((1 - 0) - (0 - 1)) / (2 \times 2) = 0.5$). Thus, the selected effect size ($d_s = 0.5$) used in our a priori power analysis would be clearly invalid and only appropriate in the case that we were expecting a reverse interaction. Lastly, for an attenuated interaction (Figure S2d), the pattern of means would correspond to a $d_{int} = 0.1$ ($((1 - 0) - (0.6 - 0)) / (2 \times 2) = 0.1$). Plugging an effect size of $d_{int} = 0.1$, a type I error of $\alpha = 0.05$, and statistical power to 0.8 into G*Power for a two-sided unpaired t -test yields a total sample size of 3142, or 786 participants per group. Using the wrong estimate of the interaction effect has dire consequences for the sample size required to find the interaction.

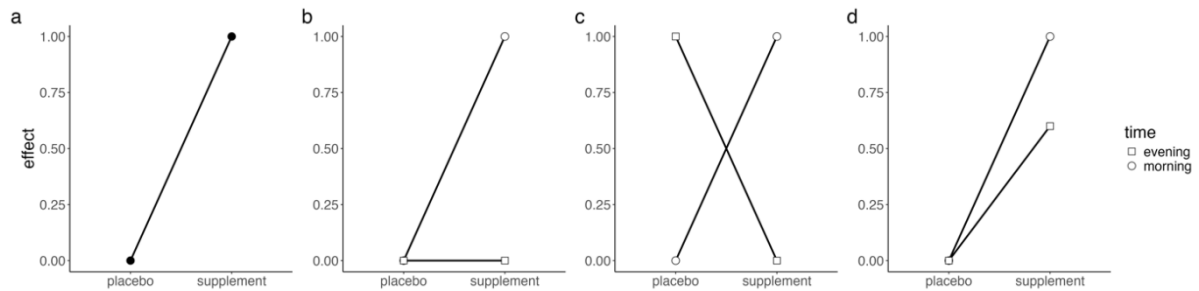


Figure S2. Illustration of a simple effect and three types of interaction. Figure S2a depicts a simple effect where researchers tested for the difference between two experimental conditions. Figure S2b depicts an ordinal interaction where intake time suppresses the effect of the supplement. Figure S2c depicts a disordinal interaction where intake time reverses the effect of the supplement. Figure S2d depicts an attenuated interaction where intake time attenuates the effect of the supplement.

As previously mentioned, because d_{int} boils down to standardised mean difference calculated as the difference between two simple standardised effect sizes (d_s), a priori power analyses can be performed using an unpaired t -test as the statistical test and d_{int} . It is worth pointing out that the t -test approach would be mathematically

equivalent to conducting an a priori power analysis for a 2 x 2 between-subject ANOVA as the statistical test to detect an interaction effect size $f = d_{int} / 2$ in G*Power.

2.2. Two mixed factorial design or two within-subject ANOVA

The difference-in-differences approach can be extended to mixed factorial designs and within-subject designs (Langenberg et al., 2023; Sommet et al., 2023) where the same principle still holds. That is, d_{int} boils down to a standardised mean difference for which a priori power analyses can be conducted using a t -test. The key difference is that the correlation between measurements must be considered. “Int×Power” is a user-friendly web app that allows researchers not only to easily estimate their expected two-way between-subject interaction and required total sample size to achieve the desired level of power, but also to estimate their expected interaction in 2 x 2 mixed designs assuming a conservative correlation estimate = 0.5 (Sommet et al., 2023).

To estimate an interaction effect for a 2 x 2 within-subject factorial design, in addition to the four cell means (see Table S1 below), we need the within-subject factor variance ($\sigma^2 = SD^2$) and the correlation (ρ) between measurements. For easiness, we will assume equal variances amongst 4 conditions ($\sigma^2 = 2^2$) as well as a common correlation ($\rho = 0.8$). For a detailed tutorial about how to calculate an interaction effect from a 2 x 2 within-subject design using different variances, readers are referred to (Langenberg et al., 2023).

Table S1. Cell means in a 2 x 2 within-subject design.

		Factor A	
		1	2
Factor B	1	10	9
	2	9	9

To calculate the interaction effect size, we first need to compute the difference of differences as follows:

$$\begin{aligned}
 \text{diff}_{A:B} &= (A1B1 - A1B2) - (A2B1 - A2B2) \\
 &= 10 - 9 - 9 + 9 \\
 &= 1
 \end{aligned} \tag{4}$$

We then need to compute the covariance, which in a 2 x 2 within-subject design can be computed as:

$$\begin{aligned}
 \sigma_{diff_{A:B}}^2 &= 2^k \cdot \sigma^2 - 2^k \cdot \sigma_{cov} \\
 \sigma_{diff_{A:B}}^2 &= 2^2 \cdot 2^2 - 2^2 \cdot (\sigma^2 \cdot \rho) \\
 \sigma_{diff_{A:B}}^2 &= 4 \cdot 4 - 4 \cdot (4 \cdot 0.8) = 3.2
 \end{aligned} \tag{5}$$

Once we have calculated $\text{diff}_{A:B}$ and $\sigma_{diff_{A:B}}^2$ we can calculate Cohen’s d_{int} as:

$$d_{int} = \frac{\text{diff}_{A:B}}{\sqrt{\sigma_{diff_{A:B}}^2}} \tag{6}$$

$$d_{int} = \frac{1}{\sqrt{3.2}} = 0.56$$

Plugging an effect size of $d_{int} = 0.56$, a type I error of $\alpha = 0.05$, and a desired level of statistical power of 0.8 into G*Power yields a total sample size of 28 participants for a two-sided paired t -test. This t -test approach would also be mathematically equivalent to conducting an a priori power analysis for a 2 x 2 within-subject ANOVA as the statistical test to detect an interaction effect size $f = d_{int}$ in G*Power. Alternatively, if the standardized mean difference between A1B1 and A1B2, and between A2B1 and A2B2 were reported as d_z , researchers could plug the average of these two d_z in G*Power to estimate the interaction effect size (Brysbaert, 2019).

3. References

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